SYRINE SMATI

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EDUCATION

National Institute of Applied Science and Technology (INSAT) — Sept 2024 – Present
Engineering Cycle in Software Engineering — Expected Degree: 2027

National Institute of Applied Science and Technology (INSAT) — Sept 2022 – June 2024
Preparatory Cycle in Mathematics, Physics, and Computer Science

Lycée Les Pères Blancs — 2022
Baccalaureate in Mathematics — **Highest Honors (Très Bien)**
Grade: 18.73/20 — Ranked 1st in Tunis 2 Region

PROFESSIONAL EXPERIENCE

Welyne – AI Intern, Ariana, Tunisia — Jul 2025 – Sep 2025

- Improved a deep learning model predicting arm length from biometric features (gender, height, weight).
- Increased accuracy to **79%** and significantly reduced MAE/MSE through preprocessing and hyperparameter tuning.
- Designed and evaluated **3 architectures** validated with cross-validation.
- **Technologies:** Python, TensorFlow, scikit-learn, NumPy, Pandas

Yonnov'IA – AI Intern, Remote (Marseille, France) — Jun 2025 – Aug 2025

- Built a modular recommendation system for Odoo eCommerce combining collaborative and content-based filtering.
- Improved recommendation coverage by **+30%**.
- Reduced cold-start problem using synthetic data generation.
- Evaluated system on **1000+ products**.
- **Technologies:** Python, Odoo, scikit-learn, Pandas

PROJECTS

GitHub Trends Analyzer – Big Data & Real-Time Analytics Pipeline

- Designed an end-to-end data pipeline combining batch processing (Airflow, PySpark) and streaming (Kafka, Spark Streaming).
- Built dashboards with Streamlit for real-time visualization and prediction.
- **Technologies:** Python, Airflow, PySpark, Kafka, HDFS, HBase, Streamlit

Tunisian Arabic Spoken Dialogue System – ASR & LLM

- Built a voice assistant for Tunisian dialect.
- Fine-tuned ASR models (Whisper, w2v-BERT, MMS) on 400h of speech data.
- Adapted LLM (Aya-8B) using QLoRA and RAG for natural responses.
- **Technologies:** PyTorch, Hugging Face, Whisper, QLoRA, RAG

DinePilot – Computer Vision Analytics Platform

- Developed real-time computer vision system for restaurant analytics (YOLO models).
- Estimated table occupancy, service time, and customer waiting time.
- Built dashboards and automated reporting.
- **Award:** 2nd Place – Smart Service Challenge (AI Camera Hackathon)
- **Technologies:** Python, YOLOv8/11, OpenCV, React

SKILLS

Machine Learning: Scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM, Feature Engineering, Cross-validation
Deep Learning: CNNs, RNNs, Transformers, LLMs, Hugging Face, Fine-tuning
Big Data: SQL, Spark, Kafka, Airflow, HDFS, HBase
Computer Vision: OpenCV, YOLO, Image Classification, Object Detection
Fullstack: React, Next.js, NestJS, FastAPI, .NET, Odoo
DevOps: Docker, Kubernetes, CI/CD, Terraform, Pulumi
Databases: PostgreSQL, MySQL, MongoDB, Redis, Neo4j
Languages: English (Fluent), French (Native), Arabic (Native), Spanish (Beginner)